

Tail Absorption Shortens the Missing Weighted Bridge Clock

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Abstract

The projection-free spectral tail was instantiated with the convenient cut $m_\sigma = \lceil 4 \log(1/\sigma) \rceil$, and the missing determinant leaf was therefore recorded as a weighted coefficient bridge up to that order. We show that this is longer than necessary. Split the prefix again at $h_\sigma = \lceil a_* \log(1/\sigma) \rceil$, where $a_* = 1/\log(10/7)$. The intervening noisy-complement coefficients are absorbed by the sharp critical mass-and-cap tail, while the deterministic coefficients are absorbed by the certified target envelope. Hence a direct bridge only to h_σ , or the two typed constituent bridges to that clock, implies the original slope-four prefix. This is a strict reduction of the reopening input, not a proof that the bridge is present.

1 Budgets

Fix $R = 7/5$, $q = 1/2$, and let

$$m_\sigma = \lceil 4L_\sigma \rceil, \quad h_\sigma = \lceil a_* L_\sigma \rceil, \quad L_\sigma = \log(1/\sigma), \quad a_* = \frac{1}{\log(1/(qR))}.$$

For complement traces $\tau_{\sigma,n}$ and target anchors a_n , write

$$P_\sigma(u) = \sum_{2 \leq n < u} \frac{|\tau_{\sigma,n} - a_n| R^n}{n}.$$

The RH-282 mass law gives $|\tau_{\sigma,n}| \leq \sigma^{-1} q^{n-2}$, and RH-267 gives $|a_n| < 48q_*^n$ with $q_* = (0.85\lambda)^{-1} = 0.7008752258 \dots$

2 Clock-shortening theorem

Theorem 2.1. *For all sufficiently small σ ,*

$$P_\sigma(m_\sigma) \leq P_\sigma(h_\sigma) + S_\sigma(h_\sigma) + T_\sigma(h_\sigma),$$

where

$$S_\sigma(h) = \sum_{n \geq h} \frac{|\tau_{\sigma,n}| R^n}{n}, \quad T_\sigma(h) = \sum_{n \geq h} \frac{|a_n| R^n}{n}.$$

Moreover,

$$S_\sigma(h_\sigma) \leq \frac{40}{3h_\sigma}, \quad T_\sigma(h_\sigma) \leq \frac{48(q_*R)^{h_\sigma}}{h_\sigma(1 - q_*R)} \longrightarrow 0.$$

Consequently $P_\sigma(h_\sigma) \rightarrow 0$ implies $P_\sigma(m_\sigma) \rightarrow 0$. It is also sufficient that the RH-288 typed budgets $E_\sigma(h_\sigma)$ and $D_\sigma(h_\sigma)$ both tend to zero.

Proof. The triangle inequality on the slab $h_\sigma \leq n < m_\sigma$ gives the first display. Since $qR = 7/10$ and $a_* \log(10/7) = 1$,

$$\sigma^{-1}(qR)^{h_\sigma} \leq \sigma^{-1}(qR)^{a_* L_\sigma} = 1.$$

Summing the geometric trace envelope gives

$$S_\sigma(h_\sigma) \leq \frac{\sigma^{-1} q^{-2} (qR)^{h_\sigma}}{h_\sigma(1 - qR)} \leq \frac{40}{3h_\sigma}.$$

The target estimate follows similarly from $q_* R < 1$. Finally, $P_\sigma(h) \leq E_\sigma(h) + D_\sigma(h)$ is the exact typed decomposition of RH-288. $\square$

3 Protocol and boundary

The accompanying computation evaluates the two certified tail bounds at five noise scales and records the critical and slope-four clocks. It tests only the displayed inequalities; no noisy coefficient data are fitted.

Remark 3.1. The theorem changes the required bridge horizon from slope 4 to the sharp mass-and-cap slope $a_* = 2.803673252\dots$. It does not prove $P_\sigma(h_\sigma) \rightarrow 0$, identify the modulus head with a counterloop, or construct a physical Riesz quotient. Gates A–E remain false/open; no Hilbert–Polya operator, Riemann-zero identification, zeta-divisor equality, von Mangoldt trace, or RH conclusion is asserted.